

Problem

Given $h(t) = 4e^{-2t}u(t)$ and $x(t) = \delta(t) - 2e^{-2t}u(t)$, find $y(t) = x(t) * h(t)$.

Step-by-step solution

Step 1 of 3

Consider the following expressions:

$$h(t) = 4e^{-2t}u(t)$$

$$x(t) = \delta(t) - 2e^{-2t}u(t)$$

Find the response $y(t)$.

Write the expression for $y(t)$.

$$\begin{aligned} y(t) &= x(t) * h(t) \\ &= [\delta(t) - 2e^{-2t}u(t)] * [4e^{-2t}u(t)] \end{aligned}$$

$$y(t) = \{\delta(t) * [4e^{-2t}u(t)]\} - \{2e^{-2t}u(t) * 4e^{-2t}u(t)\} \dots (1)$$

Refer Table 15.1 in textbook for the properties of the Laplace transform.

$$f_1(t) * f_2(t) = F_1(s)F_2(s)$$

Calculate the expression $\delta(t) * [4e^{-2t}u(t)]$.

Apply Convolution property to this expression.

$$\begin{aligned} \delta(t) * [4e^{-2t}u(t)] &= L^{-1} \left\{ (L[\delta(t)])(L[4e^{-2t}u(t)]) \right\} \\ &= L^{-1} \left\{ (1) \left(\frac{4}{s+2} \right) \right\} \\ &= 4L^{-1} \left\{ \frac{1}{s+2} \right\} \\ &= 4e^{-2t}u(t) \end{aligned}$$

Therefore, $\delta(t) * [4e^{-2t}u(t)] = 4e^{-2t}u(t)$.

Step 2 of 3

Calculate the expression $2e^{-2t}u(t) * 4e^{-2t}u(t)$.

Apply Convolution property to this expression.

$$\begin{aligned} 2e^{-2t}u(t) * 4e^{-2t}u(t) &= L^{-1} \left\{ (L[2e^{-2t}u(t)])(L[4e^{-2t}u(t)]) \right\} \\ &= L^{-1} \left\{ \left(\frac{2}{s+2} \right) \left(\frac{4}{s+2} \right) \right\} \\ &= 8L^{-1} \left\{ \frac{1}{(s+2)^2} \right\} \\ &= 8te^{-2t}u(t) \end{aligned}$$

Therefore, $2e^{-2t}u(t) * 4e^{-2t}u(t) = 8te^{-2t}u(t)$

Step 3 of 3

Substitute $8te^{-2t}u(t)$ for $2e^{-2t}u(t)*4e^{-2t}u(t)$, $4e^{-2t}u(t)$ for $\delta(t)*[4e^{-2t}u(t)]$ in this expression.

$$\begin{aligned}y(t) &= 4e^{-2t}u(t) - 8te^{-2t}u(t) \\ &= 4(1 - 2t)e^{-2t}u(t)\end{aligned}$$

Therefore, the required expression for $y(t)$ is $\boxed{4(1 - 2t)e^{-2t}u(t)}$.